Microeconomics In-class Quiz 4
Spring 2026

Student ID:

Name:

Instructions

1. Do NOT flip over this page until every student receives this quiz. Your TA will let you know when you can start.
2. During this closed-book quiz, you cannot consult any materials.
3. It is okay to write your reasoning in Korean.
4. There are five questions in total. Both sides are printed, so make sure you check the backside of each page.
5. [**IMPORTANT**] Make your answers **legible**. Clearly delineate your scratches from your answers. Deducted points due to illegible writing cannot be the reason for reevaluation.

Honor Code: Cheating on exams or quizzes, plagiarizing someone else's answers as one's own, or any other instance of academic dishonesty violates the standards of academic integrity.

Confidentiality Code: Sharing the information of the exam or quiz contents with other students in any form and medium is strongly prohibited, as it raises information inequity.

I, ________________, consent to the Honor Code and the Confidentiality Code.
(write your name)

1. AMD and Intel form a duopoly in a specialized microchip market. The market inverse demand is given by $P = 120 - Q$, where P is the market price of a chip and $Q = q_A + q_I$ is the total market quantity produced in a year. AMD's marginal cost is 20, and Intel's marginal cost is 40.

(a) [1 point] Given Intel's production level q_I, derive the best response function for AMD.

(b) [2 points] Describe the Nash equilibrium if both firms choose their output levels simultaneously. What is the total aggregate quantity in this equilibrium?

(c) [2 points] Suppose Intel is a Stackelberg leader and decides its output level first, followed by AMD. What would be the aggregate quantity produced in the Subgame-Perfect Equilibrium?

(d) [1 point] If Intel shuts down the business, and AMD forms a monopoly, what would be the profit-maximizing output level in this market?

2. A firm's short-run production function is given by $Q = 4\sqrt{L}$, and the associated marginal product of labor is $MP_L = \frac{2}{\sqrt{L}}$.

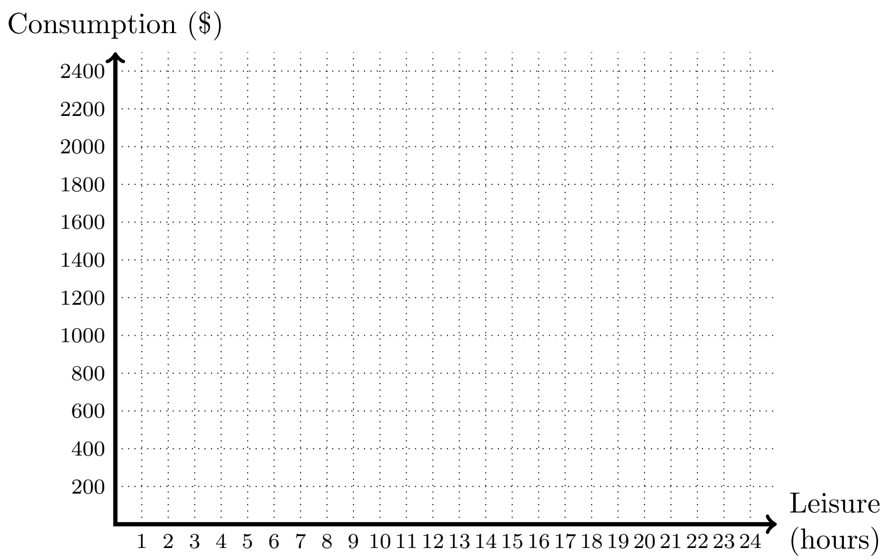
(a) [1 point] If the firm faces a constant marginal revenue of 50, what is the marginal revenue product of labor?

(b) [2 points] If the firm can hire workers at a competitive market wage w, derive the firm's labor demand function (how many workers the firm will demand as a function of w).

3. Suppose Marcus is deciding how many hours to work each day. The labor market is perfectly competitive, and his hourly wage is \$10. He can choose to work anywhere from 0 to 24 hours per day, with the remainder being his leisure hours. His entire labor income is spent on consumption.

(a) [1 point] If Marcus works for h hours, his income is $10h$, and his leisure hours are $24 - h$. Draw his budget constraint on a leisure–consumption plane.

(b) [2 points] Marcus's utility is given by $U = \min\{C, 5L\}$, where C is Consumption and L is Leisure. Draw two or more indifference curves on your graph. What is his optimal number of working hours?



4. Diana is a freelance consultant. Her utility function over income is $U = \sqrt{I}$, where I is her weekly income. In a standard week, Diana earns \$2500. However, there is a 20% chance each week that her computer equipment breaks down, requiring a sudden repair fee of \$900.

(a) [1 point] Calculate Diana's expected income and expected utility.

(b) [1 point] Suppose Diana is offered a long-term corporate contract that completely eliminates the risk of these sudden expenses. What guaranteed weekly salary would provide Diana with the same utility she expects to receive as a freelancer?

(c) [2 points] Suppose instead that Diana can buy an "IT Support Insurance" plan that fully covers any repair costs. What is the maximum premium Diana would be willing to pay to obtain this insurance?

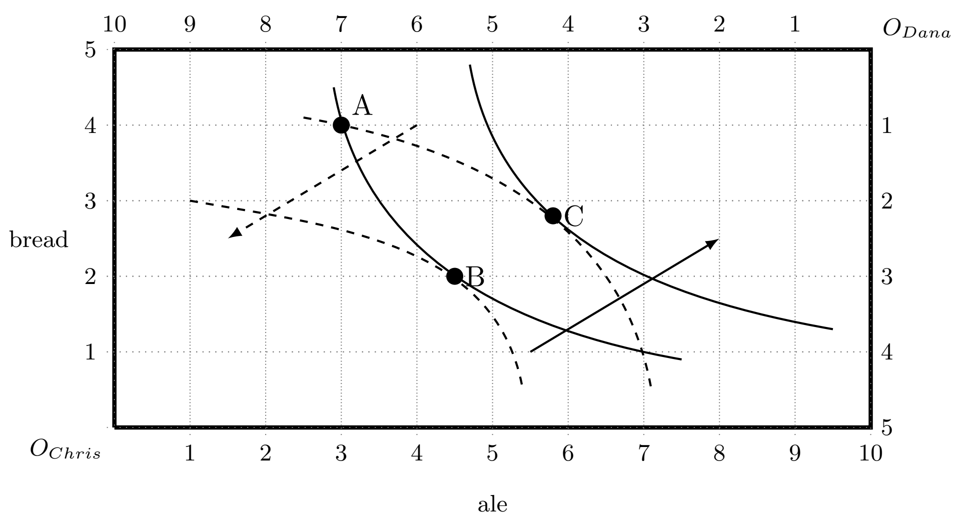
5. A colleague offers to play the following game with you: You pay a participation fee of p, and they toss a fair coin. If it lands on heads, they pay you \$20; if it lands on tails, they pay you \$60

(a) [1 point] If you are risk-neutral, what is the maximum participation fee you would be willing to pay to play?

(b) [2 points] Your colleague, Victor, has a utility function of $U = W^2$. Assuming Victor makes decisions based on expected utility, would he choose to play this game if the participation fee is \$42? (Assume his utility of not playing is exactly the utility of keeping the \$42).

(c) [1 point] Based on his utility function, is Victor risk-averse, risk-neutral, or risk-loving?

6. The Edgeworth box below describes an exchange economy of Chris and Dana with two goods—ale and bread. Arrows indicate the direction of utility increases. Dashed lines are for Dana.



(a) [1 point] Suppose the resources are allocated at point A. How many units of ale and bread do Chris and Dana have?

(b) [2 point] Is moving from point A to point C Pareto improvement? Explain.

(c) [2 point] Is moving from point B to point C Pareto improvement? Explain.